



Final Requirement Submission

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Advanced Web Development

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Abstract

This pdf is the collective work of Pawan Lamsal (1751362), Sher Afgan (1751148) and Suman Neupane (1751373), done as a part of final exercise for the module Advanced Web Development under the International Masters in Software Engineering program at the University of Hildesheim, Germany.

Student Dorm Management System

The Student Dormitory Management System, a web-based application intended to support and digitalize core administrative processes in student dormitories. The system targets two user roles, students and dormitory administrators, and defines functionality for online dormitory applications, tenant profile management, digital contract handling, rent payment record management, and complaint submission and processing. The requirements include mechanisms for uploading and downloading documents such as contracts and rent payment receipts, as well as a backend process for generating rent payment history reports as a longer-running computation with progress feedback displayed on the frontend. In addition, the system provides role-based access control, dialog-based user confirmations, and router-based menu navigation.

Requirements Initial

Functional Requirements (Must)

R01: The system must provide registration functionality for students.

R02: The system must provide authentication and login functionality to students.

R02a: The system must provide authentication and login functionality to dormitory administrators.

R03: The system must provide an online dormitory application form for students.

R04: The system must provide a personal profile for students.

R05: The system must provide a dashboard for dormitory administrators.

R06: The system must provide application review functionality to dormitory administrators.

R07: The system must provide application acceptance functionality to dormitory administrators.

R07a: The system must provide application rejection functionality to dormitory administrators.

R08: The system must provide file upload functionality for students for submitting required documents.

R09: The system must provide digital contract creation for accepted students.

R10: The system must provide file upload functionality for students to upload signed rental contracts.

Explanation: Signed contracts are uploaded and stored in the system (mandatory file upload requirement).

R11: The system must provide file download functionality for students to download their uploaded contracts.

Explanation: Contracts can be downloaded as PDF files (mandatory file download requirement).

R12: The system must provide file upload functionality for students for uploading monthly rent payment receipts.

R13: The system must provide storage of rent payment records for students.

R14: The system must provide file download functionality for students to download rent payment records as a PDF document.

R15: The system must provide contract extension functionality to dormitory administrators.

R16: The system must provide contract termination functionality to dormitory administrators.

R17: The system must provide a backend process to generate a rent payment history report.
Explanation: This represents a computationally longer-running backend task (mandatory long-running task requirement).

R18: The system must provide progress status information to students during the generation of rent payment reports.
Explanation: Progress of the long-running backend task is displayed on the frontend (mandatory progress display requirement).

R19: The system must provide a visual progress indicator to students while backend processing is running.
Explanation: Visual representation of backend task progress on the frontend.

R20: The system must provide complaint submission functionality for students.

R21: The system must provide complaint management functionality to dormitory administrators.

R22: The system must provide a dialog window for students to confirm complaint submission.
Explanation: This fulfills the mandatory requirement for at least one meaningful dialog window (not related to file upload/download).

R23: The system must provide a dialog window for dormitory administrators to confirm application acceptance or rejection.

R24: The system must provide menu-based navigation for all users using Angular routing.
Explanation: This fulfills the mandatory router-based menu requirement.

R25: The system must provide role-based access control for students.

R25a: The system must provide role-based access control for dormitory administrators.

Functional Requirements (Should)

R26: The system shall provide an overview dashboard for students showing application status.

R26a: The system shall provide an overview dashboard for students showing contract status.

R27: The system shall provide an overview dashboard for dormitory administrators showing pending applications.

R28: The system shall provide the ability to view complaint status for students.

R29: The system shall provide the ability to filter rent payments by month for students.

R30: The system shall provide basic validation of uploaded documents for students.

Explanation: File type and size validation for uploaded documents.

R31: The system shall provide notification messages for students after successful actions.

Functional Requirements (Can)

R32: The system can provide search functionality for dormitory administrators for tenant records.

R33: The system can provide data export functionality for dormitory administrators.

R34: The system can provide profile update functionality for students.

Mandatory Technical / Acceptance Requirements (Not Counted)

TR01: The system must be implemented using an Angular frontend with router-based navigation.

TR02: The system must utilize backend technology to support the frontend.

TR03: The system must provide automated tests executed with Karma.

TR04: The system must achieve test coverage above 60%, including e2e tests.

TR05: The system must be delivered as a runnable VirtualBox VM including documentation until 2026/03/17.